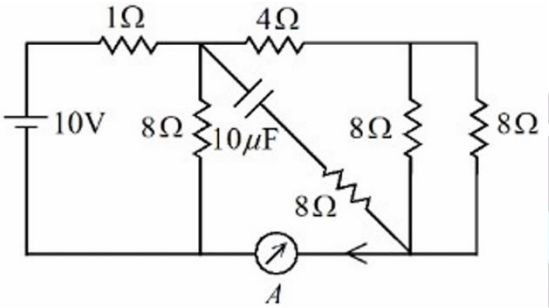
24 January 2026, Shift 2

1. The reading of the ammeter (A) in steady state in the following circuit (assuming negligible internal resistance of the ammeter) is_____A.



a. 2

b. 1

c. 1/2

d. 0

2. In a vernier callipers, 50 vernier scale divisions are equal to 48 main scale divisions. If one main scale division = 0.05 mm, then the least count of the vernier callipers is ___ mm.

a. 0.002

b. 0.05

c. 0.02

d. 0.005

3. Five persons P_1, P_2, P_3, P_4 and P_5 recorded object distance (u) and image distance (v) using same convex lens having power +5 D as $(25,96),(30,62),(35,37),(45,35)$ and $(50,32)$ respectively. Identify correct statement

a. Readings recorded by all persons are correct

b. Readings recorded by P_3 person are incorrect

c. Readings recorded by P_3 and P_2 persons are incorrect

d. Readings recorded by P_4 and P_5 persons are incorrect

4. A regular hexagon is formed by six wires each of resistance $r\Omega$ and the corners are joined to the centre by wires of same resistance. If the current enters at one corner and leaves at the opposite corner, the equivalent resistance of the hexagon between the two opposite corners will be

a. $\frac{4}{5}r$

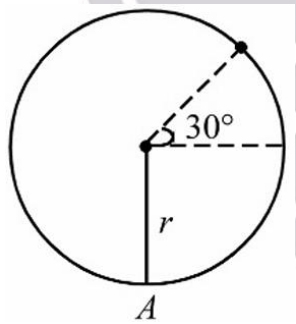
b. $\frac{5}{8}r$

c. $\frac{3}{4}r$

d. $\frac{3}{5}r$

5. In case of vertical circular motion of a particle by a thread of length r if the tension in the thread is zero at an angle 30º shown in figure, the velocity at the bottom point (A) of the circular path is

(g = gravitational acceleration)



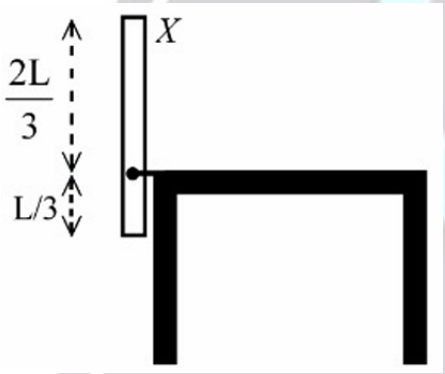
a. $\sqrt{5gr}$

b. $\sqrt{\frac{7}{2}gr}$

c. $\sqrt{4gr}$

d. $\sqrt{\frac{5}{2}gr}$

6. A thin uniform rod (X) of mass M and length L is pivoted at a height $\left(\frac{L}{3}\right)$ as shown in the figure. The rod is allowed to fall from a vertical position and lie horizontally on the table. The angular velocity of this rod when it hits the table top, is____. (g = gravitational acceleration)



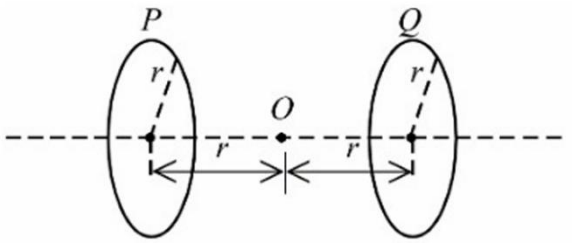
a. $\sqrt{\frac{3}{2}\frac{g}{L}}$

b. $\frac{3}{\sqrt{2}}\sqrt{\frac{g}{L}}$

c. $\frac{1}{\sqrt{2}}\sqrt{\frac{g}{L}}$

d. $\sqrt{\frac{3g}{L}}$

7. Two identical circular loops P and Q each of radius r are lying in parallel planes such that they have common axis. The current through P and Q are I and 4I respectively in clockwise direction as seen from O. The net magnetic field at O is:



a. $\frac{3\mu_0 I}{4\sqrt{2}r}$ towards P

b. $\frac{\mu_0 I}{4\sqrt{2}r}$ towards P

c. $\frac{\mu_0 I}{4\sqrt{2}r}$ towards Q

d. $\frac{3\mu_0 I}{4\sqrt{2}r}$ towards Q

8. When a light of a given wavelength falls on a metallic surface the stopping potential for photoelectrons is 3.2 V. If a second light having wavelength twice of first light is used, the stopping potential drops to 0.7 V. The wavelength of first light is_____m.

$\left(h = 6.63\times10^{-34}\ \text{J.s}, e = 1.6\times10^{-19}\ \text{C}, c = 3\times10^{8}\ \text{m/s}\right)$

a. 2.9×10^{-8}

b. 2.2×10^{-8}

c. 3.1×10^{-7}

d. 2.5×10^{-7}

9. The fifth harmonic of a closed organ pipe is found to be in unison with the first harmonic of an open pipe. The ratio of lengths of closed pipe to that of the open pipe is $5/x$. The value of x is ——— .

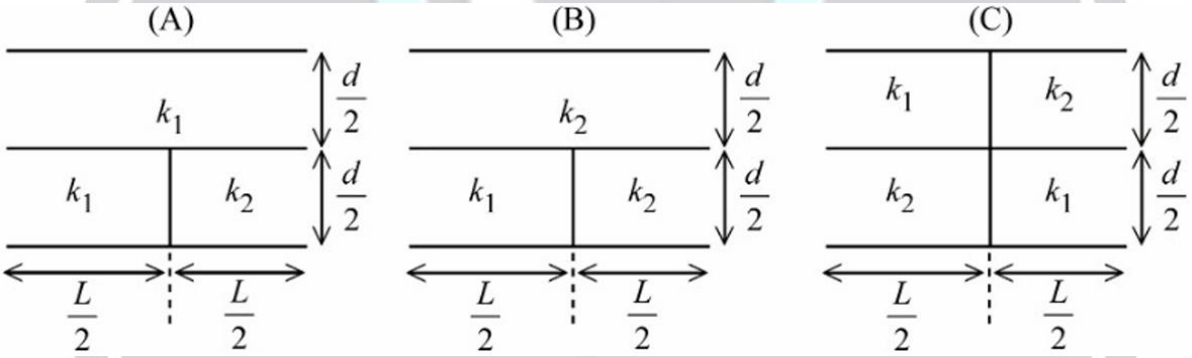
a. 4

b. 2

c. 1

d. 3

10. Three parallel plate capacitors each with area A and separation d are filled with two dielectric (k_1 and k_2) in the following fashion. Which of the following is true?



a. $C_B > C_C > C_A$

b. $C_C > C_B > C_A$

c. $C_C > C_A > C_B$

d. $C_A > C_C > C_B$

11. The binding energy for the following nuclear reactions are expressed in MeV .

$$_2He^3 +_0 n^1 \rightarrow_2 He^4 + 20MeV$$

$_2He^4 +_0 n^1 \rightarrow_2 He^5 - 0.9MeV$

If X_3, X_4, X_5 denote the stability of $_2He^3$, $_2He^4$ and $_2He^5$, respectively, then the correct order is :

a. $X_4 > X_5 > X_3$

b. $X_4 = X_5 = X_3$

c. $X_4 > X_5 < X_3$

d. $X_4 < X_5 < X_3$

12. A cubical block of density $\rho_b = 600\,kg/m^3$ floats in a liquid of density $\rho_c = 900kg/m^3$. If the height of block is $H = 8.0\,cm$ then height of the submerged part is —— cm .

a. 7.3

b. 4.3

c. 6.3

d. 5.3

13. A moving coil galvanometer of resistance 100Ω shows a full scale deflection for a current of 1 mA . The value of resistance required to convert this galvanometer into an ammeter, showing full scale deflection for a current of 5 mA , is —— Ω

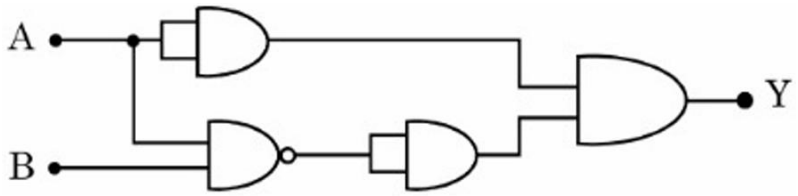
a. 25

b. 10

c. 0.5

d. 2.5

14. Identify the correct truth table of the given logic circuit.



a.

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

b.

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	0

c.

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

d.

A	B	Y
0	0	0
0	1	0
1	0	1
1	1	0

15. A flexible chain of mass m hangs between two fixed points at the same level. The inclination of the chain with the horizontal at the two points of support is 30º. Considering the equilibrium of each half of the chain, the tension of the chain at the lowest point is ——— .

a. $\frac{\sqrt{3}}{2}mg$

b. $\frac{1}{2}mg$

c. mg

d. $\sqrt{3}\text{mg}$

16. A point source is kept at the center of a spherically enclosed detector. If the volume of the detector increased by 8 times, the intensity will

a. increase by 8 times

b. increase by 64 times

c. decrease by 8 times

d. decrease by 4 times

17. In the Young's double slit experiment the intensity produced by each one of the individual slits is I_o . The distance between two slits is 2 mm . The distance of screen from slits is 10 m . The wavelength of light is $6000\ \text{Å}.$ The intensity of light on the screen in front of one of the slits is ——

a. $2I_o$

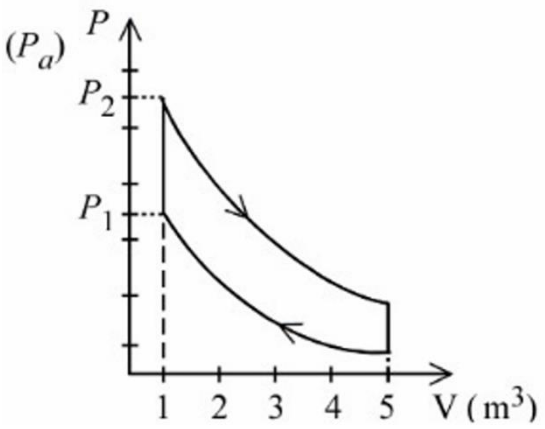
b. I_o

c. $\frac{I_o}{2}$

d. $4I_o$

18. 10 mole of an ideal gas is undergoing the process shown in the figure. The heat involved in the process from P_1 to P_2 is α Joule

($P_1 = 21.7\,\text{Pa}$ and $P_2 = 30\,\text{Pa}, C_v = 21\,\text{J/K.mol}, R = 8.3\,\text{J/mol.K}$). The value of α is ——— .



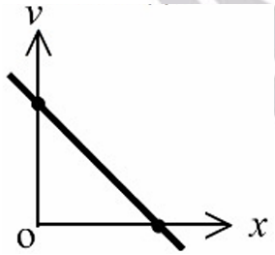
a. 24

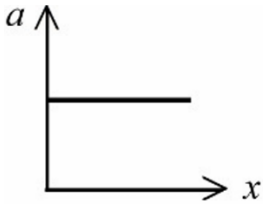
b. 15

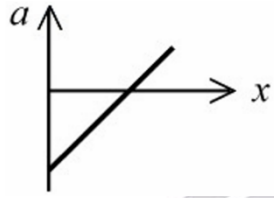
c. 21

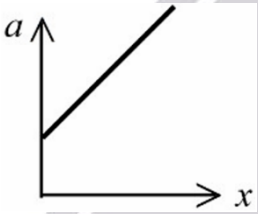
d. 28

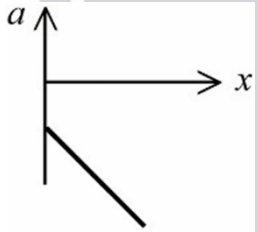
19. The velocity (v) - Distance (x) graph is shown in figure. Which graph represents acceleration(a) versus distance (x) variation of this system?



a.


b.


c.


d.


20. Distance between an object and three times magnified real image is 40 cm. The focal length of the mirror used is _____cm.

a. –15 / 2

b. -10

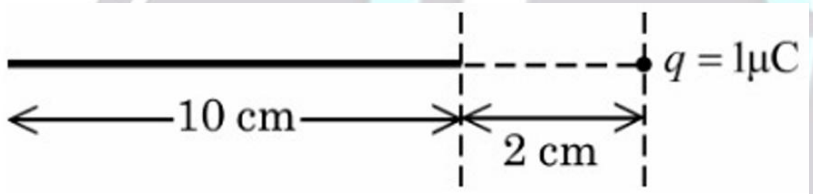
c. -20

d. -15

$C_p = \frac{7}{2}R$

21. When 300 J of heat given to an ideal gas with its temperature raises from $20^\circ C$ to $50^\circ C$ keeping its volume constant. The mass of the gas is (approximately) ____ g. $(R = 8.314\,J/mol.K)$

22. A point charge $q = 1\mu C$ is located at a distance 2 cm from one end of a thin insulating wire of length 10 cm having a charge $Q = 24\mu C$, distributed uniformly along its length, as shown in figure. Force between q and wire is ____ N. $(Use: \frac{1}{4\pi\grave{o}_o} = 9\times10^9\ N\cdot m^2/C^2)$



90

23. In a meter bridge experiment to determine the value of unknown resistance, first the resistances 2Ω and 3Ω are connected in the left and right gaps of the bridge and the null point is obtained at a distance $l\,cm$ from the left. Now when an unknown resistance $x\Omega$ is connected in parallel to 3Ω resistance, the null point is shifted by 10 cm to the right of wire. The value of unknown resistance x is____ Ω.

24. A uniform solid cylinder of length L and radius R has moment of inertia about its axis equal to I_1. A small co-centric cylinder of length L/2 and radius R/3 carved from this cylinder has moment of inertia about its axis equals to I_2. The ratio I_1/I_2 is ____ .

25. A soap bubble of surface tension $0.04\,\mathrm{N/m}$ is blown to a diameter of 7 cm . If $(15000-x)\mu\mathrm{J}$ of work is done in blowing it further to make its diameter 14 cm, then the value of x is _____. $(\pi = 22/7)$

26. Choose the INCORRECT statement

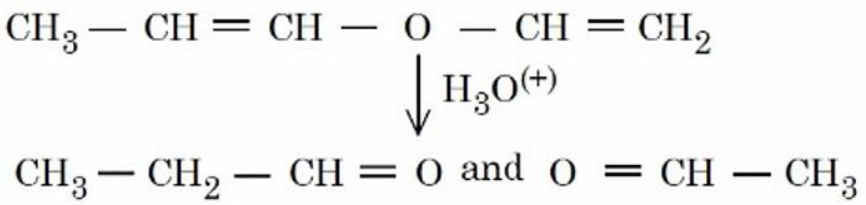
a. Among the isotopes of carbon, $^{13}\mathrm{C}$ is a radioactive isotope.

b. Carbon exhibits negative oxidation states along with +4 and +2 .

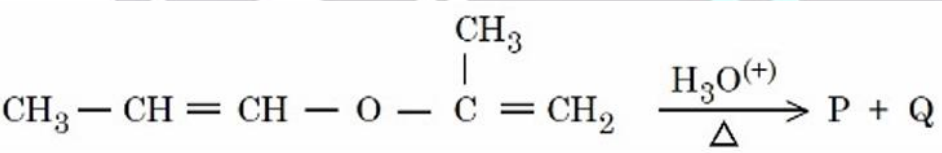
c. Carbon cannot exceed its covalency more than four.

d. $\mathrm{CO_2}$ is the most acidic oxide among the dioxides of group of 14 elements.

27. The unsaturated ether on acidic hydrolysis produces carbonyl compounds as shown below:-



Based on this, predict the solution/reagent that will help to distinguish " P " and " Q " obtained in the following reaction:-



a. Lucas reagent

b. 2, 4 - DNP reagent

c. Saturated $\mathrm{NaHSO_3}$ solution

d. Fehling solution

28. The number of possible tripeptides formed involving alanine (ala), glycine (gly) and valine (val), where no amino acid has been used more than once is:

a. 6

b. 3

c. 4

d. 8

29. Two liquids A and B form an ideal solution at temperature T K . At T K , the vapour pressures of pure A and B are 55 and $15\mathrm{kNm}^{-2}$ respectively. What is the mole fraction of A in solution of A and B in equilibrium with a vapour in which the mole fraction of A is 0.8 ?

a. 0.5217

b. 0.480

c. 0.663

d. 0.340

30. Consider the following gaseous equilibrium in a closed container of volume ' V ' at $\mathrm{T(K)}$.

2 moles each of $\mathrm{P_2(g), Q_2(g)}$ and $\mathrm{PQ(g)}$ are present at equilibrium. Now one mole each of ' $\mathrm{P_2}$ ' and ' $\mathrm{Q_2}$ ' are added to the equilibrium keeping the temperature at $\mathrm{T(K)}$. The number of moles of $\mathrm{P_2, Q_2}$ and PQ at the new equilibrium, respectively, are

a. 2.67,2.67,2.67

b. 1.21, 2.24, 1.56

c. 1.66, 1.66, 1.66

d. 2.56, 1.62, 2.24

31. Pair of species among the following having same bond order as well as paramagnetic character will be-

a. O_2^+, N_2^{2-}

b. O_2^-, N_2^+

c. O_2^+, N_2^-

d. O_2^-, N_2^-

32. The wavelength of light absorbed for the following complexes are in the order

$[Co(NH_3)_6]^{3+}$; $[Co(H_2O)_6]^{3+}$; $[Co(CN)_6]^{3-}$; $[Co(NH_3)_5(H_2O)]^{3+}$; $[CoF_6]^{3-}$

(I) (II) (III) (IV) (V)

a. III < I < II < IV < V

b. III < I < IV < V < II

c. III < IV < I < II < V

d. III < I < IV < II < V

33. One mole of $Cl_2(g)$ was passed into 2 L of cold 2 M KOH solution. After the reaction, the concentrations of Cl^-, ClO^- and OH^- are respectively (assume volume remains constant)

a. $0.75M, 0.75M, 1M$

b. $0.5M, 0.5M, 0.5M$

c. $0.5M, 0.5M, 1M$

d. $1M, 1M, 1M$

34. A student has planned to prepare acetanilide from aniline using acetic anhydride.

The student has started from 9.3 g of aniline. However, the student has managed to obtain 11 g of dry acetanilide.

The % yield of this reaction is :-

a. 81.5%

b. 97.5%

c. 59.5%

d. 72.5%

35. Find out the statements which are not true.

A. Resonating structures with more number of covalent bonds and lesser charge separation are more stable.

B. In electromeric effect, an unsaturated system shows +E effect with nucleophile and -E effect with electrophile.

C. Inductive effect is responsible for high melting point, boiling point and dipole moment of polar compounds.

D. The greater the number of alkyl groups attached to the doubly bonded carbon atoms, higher is the heat of hydrogenation.

E. Stability of carbanion increases with the increase in $\mathrm{S}-$ character of the carbon carrying the negative charge.

Choose the correct answer from the options given below:

a. A, D & E only

b. B, D & E only

c. A, C & D only

d. B & D only

36. The correct order of $\mathrm{C, N, O}$ and F in terms of second ionisation potential is

a. $\mathrm{F < N < C < O}$

b. $C < O < N < F$

c. $C < N < F < O$

d. $C < F < N < O$

37.

In the Group analysis of cations, Ba^{2+} & Ca^{2+} are precipitated respectively as

a.

sulphide & sulphide

b. hydroxide & carbonate

c. carbonate & carbonate

d. chromate & sulphide

38. The wavelength of spectral line obtained in the spectrum of Li^{2+} ion, when the transition takes place between two levels whose sum is 4 and difference is 2 , is

a. 2.28×10^{-7} cm

b. 2.28×10^{-6} cm

c. 1.14×10^{-7} cm

d. 1.14×10^{-6} cm

39. Given below are two statements:

Statement I: Cross aldol condensation between two different aldehydes will always produce four different products.

Statement II: When semicarbazide reacts with a mixture of benzaldehyde and acetophenone under optimum pH , it forms a condensation product with acetophenone only.

In the light of the above statements, choose the correct answer from the options given below

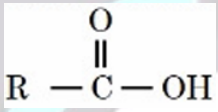
a. Both Statement I and Statement II are false

b. Statement I is false but Statement II is true

c. Both Statement I and Statement II are true

d. Statement I is true but Statement II is false

40. Given below are two statements:

Statement I: The dipole moment of R-CN is greater than R-NC and R-NC can undergo hydrolysis under acidic medium to produce  .

Statement II: R-CN hydrolyses under acidic medium to produce a compound which on treatment with $SOCl_2$, followed by the addition of NH_3 gives another compound (x). This compound (x) on treatment with $NaOCl / NaOH$ gives a product, that on treatment with $CHCl_3 / KOH / \Delta$ produces R-NC

In the light of the above statements, choose the correct answer from the options given below

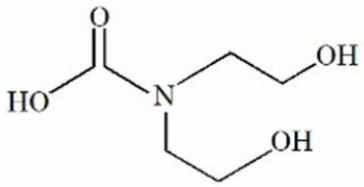
a. Both Statement I and Statement II are false

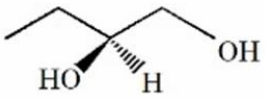
b. Both Statement I and Statement II are true

c. Statement I is true but Statement II is false

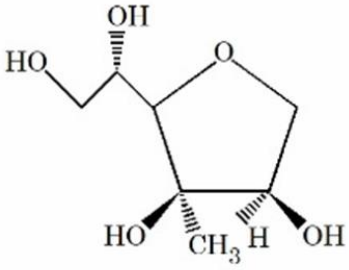
d. Statement I is false but Statement II is true

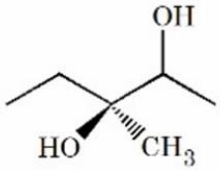
41. From the following, how many compounds contain at least one secondary alcohol?

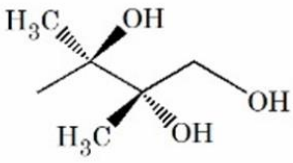

(I)


(II)


(III)


(IV)


(V)


(VI)

Choose the correct answer from the options given below:

a. Five

b. Three

c. Four

d. Two

42. The heat of atomisation of methane and ethane are ' x ' $kJmol^{-1}$ and ' y ' $kJmol^{-1}$ respectively. The longest wavelength (λ) of light capable of breaking the $C-C$ bond can be expressed in SI unit as:

a. $\frac{hc}{1000}\left(\frac{y-6x}{4}\right)^{-1}$

b. $\frac{N_A hc}{250(4y-6x)}$

c. $\frac{N_A hc}{250(y-6x)}$

d. $N_A hc\left(y-\frac{6x}{4}\right)^{-1}$

43. At 298 K , the mole percentage of $N_2(g)$ in air is 80% . Water is in equilibrium with air at a pressure of 10 atm . What is the mole fraction of $N_2(g)$ in water at 298 K ? $\left(K_H \text{ for } N_2 \text{ is } 6.5\times10^7 \text{ mmHg}\right)$

a. 1.23×10^{-7}

b. 1.17×10^{-4}

c. 9.35×10^{5}

d. 9.35×10^{-5}

44. "X" is an oxoanion of the lightest element of group 7 (in the periodic table). The metal is in +6 oxidation state in " X ". The color of the potassium salt of X is

a. Green

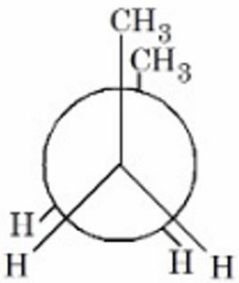
b. Purple

c. Yellow

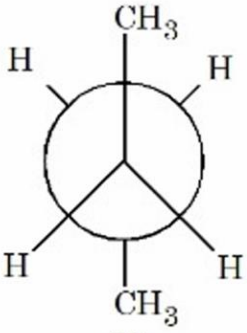
d. Orange

45. Given below are two statements:

Statement I: There are several conformers for n-butane. Out of those conformers,



(X)



(Y)

is the least stable and most stable conformer is

Statement II: As the dihedral angle increases, torsional strain decreases from (X) to (Y).

In the light of the above statements, choose the correct answer from the options given below

a. Both Statement I and Statement II are false

b. Statement I is false but Statement II is true

c. Statement I is true but Statement II is false

d. Both Statement I and Statement II are true

46. 0.25 g of an organic compound " A " containing carbon, hydrogen and oxygen was analysed using the combustion method. There was an increase in mass of $CaCl_2$ tube and potash tube at the end of the experiment. The amount was found to be 0.15 g and 0.1837 g , respectively. The percentage of oxygen in compound A is —— %. (Nearest integer)

(Given: molar mass in $gmol^{-1} H:1, C:12, O:16$)

47. The half-life of ^{65}Zn is 245 days. After x days, 75% of original activity remained. The value of x in days is —— . (Nearest integer) (Given: $\log 3 = 0.4771$ and $\log 2 = 0.3010$)

48. A chromium complex with a formula $CrCl_3 \cdot 6H_2O$ has a spin only magnetic moment value of 3.87 BM and its solution conductivity corresponds to 1: 2 electrolyte. 2.75 g of the complex solution was initially passed through a cation exchanger. The solution obtained after the process was reacted

with excess of $AgNO_3$. The amount of AgCl formed in the above process is ——— g. (Nearest integer)

[Given: Molar mass in $gmol^{-1} Cr:52; Cl:35.5, Ag:108, O:16, H:1$]

49. Molar conductivity of a weak acid HQ of concentration 0.18 M was found to be $1/30$ of the molar conductivity of another weak acid HZ with concentration of 0.02 M . If $\lambda^{\circ}Q^{-}$ happened to be equal with $\lambda^{\circ}Z^{-}$, then the difference of the pK_a values of the two weak acids $\left(pK_a(HQ) - pK_a(HZ)\right)$ is ——— (Nearest integer).

[Given: degree of dissociation for both weak acids, λ° : limiting molar conductivity of ions]

50. Grignard reagent $RMgBr(P)$ reacts with water and forms a gas (Q). One gram of Q occupies $1.4dm^3$ at STP. (P) on reaction with dry ice in dry ether followed by H_3O^+ forms a compound (Z). 0.1 mole of (Z) will weigh ——— g. (Nearest integer)

51. Let $f(x) = \int \frac{7x^{10} + 9x^8}{\left(1 + x^2 + 2x^9\right)^2} dx, x > 0, \lim_{x \to 0} f(x) = 0$ and $f(1) = \frac{1}{4}$.

If $A = \begin{bmatrix} 0 & 0 & 1 \\ \frac{1}{4} & f'(1) & 1 \\ \alpha^2 & 4 & 1 \end{bmatrix}$ and $B = \text{adj}(\text{adj}\, A)$ be such that $|B| = 81$, then α^2 is equal to

a. 2

b. 3

c. 1

d. 4

52. Let the length of the latus rectum of an ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1,(a>b)$, be 30 . If its eccentricity is the maximum value of the function $f(t)=-\frac{3}{4}+2t-t^2$, then (a^2+b^2) is equal to

a. 516

b. 256

c. 496

d. 276

53. Let the angles made with the positive x-axis by two straight lines drawn from the point $\mathrm{P}(2,3)$ and meeting the line $x+y=6$ at a distance $\sqrt{\frac{2}{3}}$ from the point P be θ_1 and θ_2. Then the value of $(\theta_1+\theta_2)$ is:

a. $\frac{\pi}{12}$

b. $\frac{\pi}{6}$

c. $\frac{\pi}{2}$

d. $\frac{\pi}{3}$

54.

Let $\vec{a} = 2\hat{i} - \hat{j} - \hat{k}, \vec{b} = \hat{i} + 3\hat{j} - \hat{k}$ and $\vec{c} = 2\hat{i} + \hat{j} + 3\hat{k}$. Let $\vec{v}$ be the vector in the plane of the vectors $\vec{a}$ and $\vec{b}$, such that the length of its projection on the vector $\vec{c}$ is $\frac{1}{\sqrt{14}}$. Then $|\vec{v}|$ is equal to

a.

$\frac{\sqrt{21}}{2}$

b. 13

c. $\frac{\sqrt{35}}{2}$

d. 7

55. Let a_1, a_2, a_3, a_4 be an A.P. of four terms such that each term of the A.P. and its common difference l are integers. If $a_1 + a_2 + a_3 + a_4 = 48$ and $a_1 a_2 a_3 a_4 + l^4 = 361$, then the largest term of the A.P. is equal to

a. 27

b. 24

c. 21

d. 23

56. Let the image of parabola $x^2 = 4y$, in the line $x - y = 1$ be $(y + a)^2 = b(x - c)$, $a, b, c \in \mathrm{N}$. Then $a + b + c$ is equal to

a. 12

b. 4

c. 6

d. 8

57. $\left(\frac{1}{3}+\frac{4}{7}\right)+\left(\frac{1}{3^2}+\frac{1}{3}\times\frac{4}{7}+\frac{4^2}{7^2}\right)+\left(\frac{1}{3^3}+\frac{1}{3^2}\times\frac{4}{7}+\frac{1}{3}\times\frac{4^2}{7^2}+\frac{4^3}{7^3}\right)+\ldots$ upto infinite terms, is equal to

a. $\frac{5}{2}$

b. $\frac{7}{4}$

c. $\frac{4}{3}$

d. $\frac{6}{5}$

58. Let $P=\left[p_{ij}\right]$ and $Q=\left[q_{ij}\right]$ be two square matrices of order 3 such that $q_{ij}=2^{(i+j-1)}p_{ij}$ and $\det(Q)=2^{10}$. Then the value of $\det(\text{adj}(\text{adj}\,P))$ is:

a. 32

b. 16

c. 81

d. 124

59. The letters of the word "UDAYPUR" are written in all possible ways with or without meaning and these words are arranged as in a dictionary. The rank of the word "UDAYPUR" is

a. 1580

b. 1578

c. 1579

d. 1581

60. The largest value of n, for which 40^n divides 60 !, is

a. 13

b. 11

c. 12

d. 14

61. The sum of all values of α, for which the shortest distance between the lines $\frac{x+1}{\alpha}=\frac{y-2}{-1}=\frac{z-4}{-\alpha}$ and $\frac{x}{\alpha}=\frac{y-1}{2}=\frac{z-1}{2\alpha}$ is $\sqrt{2}$, is

a. 8

b. -6

c. 6

d. -8

62. Let $f(\alpha)$ denote the area of the region in the first quadrant bounded by $x=0, x=1, y^2=x$ and $y=|\alpha x-5|-|1-\alpha x|+\alpha x^2$. Then $(f(0)+f(1))$ is equal to

a. 9

b. 14

c. 7

d. 12

63. If the domain of the function $f(x)=\sin^{-1}\left(\frac{1}{x^2-2x-2}\right)$, is $(-\infty,\alpha]\cup[\beta,\gamma]\cup[\delta,\infty)$, then $\alpha+\beta+\gamma+\delta$ is equal to

a. 2

b. 4

c. 3

d. 5

64.

Let $X=\{x \in N: 1 \leq x \leq 19\}$ and for some $a, b \in \mathbb{R}, Y=\{a x+b: x \in X\}$. If the mean and variance of the elements of Y are 30 and 750, respectively, then the sum of all possible values of b is

a.20

b. 80

c. 100

d. 60

65. Consider the following three statements for the function defined by $f(x)=\left|\log_e x\right|-|x-1|$:

(I) f is differentiable at all $x>0$.

(II) f is increasing in $(0,1)$.

(III) f is decreasing in $(1,\infty)$.

Then.

a. All (I), (II) and (III) are TRUE.

b. Only (I) is TRUE.

c. Only (II) and (III) are TRUE.

d. Only (I) and (III) are TRUE.

66. Let $y = y(x)$ be a differentiable function in the interval $(0, \infty)$ such that $y(1) = 2$, and $\lim\limits_{t \to x} \left(\dfrac{t^2 y(x) - x^2 y(t)}{x - t} \right) = 3$ for each $x > 0$. Then $2y(2)$ is equal to

a. 18

b. 23

c. 27

d. 12

67. Let f be a function such that $3f(x) + 2f\left(\dfrac{m}{19x}\right) = 5x, x \neq 0$, where $m = \sum\limits_{i=1}^{9} (i)^2$. Then $f(5) - f(2)$ is equal to

a. –9

b. 36

c. 18

d. 9

68. The smallest positive integral value of a, for which all the roots of $x^4 - ax^2 + 9 = 0$ are real and distinct, is equal to

a. 9

b. 3

c. 4

d. 7

69.

Let $\vec{a} = 2\hat{i} - 5\hat{j} + 5\hat{k}$ and $\vec{b} = \hat{i} - \hat{j} + 3\hat{k}$. If $\vec{c}$ is a vector such that $2(\vec{a} \times \vec{c}) + 3(\vec{b} \times \vec{c}) = \vec{0}$ and $(\vec{a} - \vec{b}) \cdot \vec{c} = -97$, then $|\vec{c} \times \hat{k}|^2$ is equal to

a. 193

b. 233

c. 218

d. 205

70. Let [t] denote the greatest integer less than or equal to t. If the function

$$f(x) = \begin{cases} b^2 \sin\left(\frac{\pi}{2}\left[\frac{\pi}{2}(\cos x + \sin x)\cos x\right]\right), & x < 0 \\ \dfrac{\sin x - \frac{1}{2}\sin 2x}{x^3}, & x > 0 \\ a, & x = 0 \end{cases}$$

is continuous at $x = 0$, then $a^2 + b^2$ is equal to

a. $\frac{5}{8}$

b. $\frac{9}{16}$

c. $\frac{3}{4}$

d. $\frac{1}{2}$

71. If $f(x)$ satisfies the relation $f(x)=e^x+\int_0^1\left(y+xe^x\right)f(y)dy$, then $e+f(0)$ is equal to ——— .

72. Let (h,k) lie on the circle $C: x^2+y^2=4$ and the point $(2h+1,3k+2)$ lie on an ellipse with eccentricity e. Then the value of $\frac{5}{e^2}$ is equal to ———

73. Let $z=(1+i)(1+2i)(1+3i)....(1+ni)$, where $i=\sqrt{-1}$. If $|z|^2=44200$, then n is equal to ———

74. Let S be a set of 5 elements and $P(S)$ denote the power set of S . Let E be an event of choosing an ordered pair (A,B) from the set $P(S)\times P(S)$ such that $A\cap B=\phi$. If the probability of the event E is $\frac{3^p}{2^q}$, where $p,q\in N$, then $p+q$ is equal to_____.

75. The number of elements in the set

$\left\{x\in\left[0,180^\circ\right]:\tan\left(x+100^\circ\right)=\tan\left(x+50^\circ\right)\tan x\tan\left(x-50^\circ\right)\right\}$ is ——— .

Answer Key

Q No.	Answer	Q No.	Answer	Q No.	Answer	Q No.	Answer
1	b	21	481	41	b	61	b
2	a	22	90	42	b	62	c
3	b	23	6	43	d	63	b
4	a	24	162	44	a	64	d
5	b	25	11304	45	d	65	d
6	d	26	a	46	73	66	b
7	d	27	d	47	102	67	c
8	d	28	a	48	3	68	d
9	b	29	a	49	2	69	c
10	d	30	a	50	6	70	c
11	a	31	c	51	d	71	2
12	d	32	d	52	c	72	9
13	a	33	c	53	c	73	5
14	d	34	a	54	c	74	15
15	a	35	d	55	a	75	4
16	d	36	c	56	c		
17	b	37	c	57	a		
18	c	38	d	58	b		
19	b	39	a	59	a		
20	d	40	d	60	d		